第十次作业答案

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1. 计算下列积分:

$$\int_V xy^2z^3\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$

V 由 $z=xy$ 和 $z=0$ 以及两张平面 $x=1$ 和 $x=y$ 围成.

解. 我们有

$$\begin{aligned}\iiint_V xy^2z^3\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z &= \int_0^1 y^2\,\mathrm{d}y\int_y^1 x\,\mathrm{d}x\int_0^{xy} z^3\,\mathrm{d}z\\ &= \int_0^1 y^2\,\mathrm{d}y\int_y^1 x\frac{x^4y^4}{4}\,\mathrm{d}x\\ &= \frac{1}{4}\int_0^1 y^6\frac{1-y^6}{6}\,\mathrm{d}y = \frac{1}{364}.\end{aligned}$$

2. 计算下列曲面围成的立体的体积:

(1). $z^2=x^2+\frac{y^2}{4}, 2z=x^2+\frac{y^2}{4}$;

(2). $x^2+z^2=a^2, |x|+|y|=a.$

解.

(1). 我们有

$$V=\int_0^2\,\mathrm{d}z\iint_{z^2\le x^2+\frac{y^2}{4}\le 2z}\,\mathrm{d}x\,\mathrm{d}y$$

对一个确定的 z, 限制区域 $z^2\le x^2+\frac{y^2}{4}\le 2z$ 为两个椭圆之间的部分, 其面积为 $2\pi\left(2z-z^2\right)$. 从而

$$V=\int_0^2 2\pi\left(2z-z^2\right)\mathrm{d}z=\frac{8}{3}\pi$$

(2). 此时

$$V=\iint_{|x|+|y|\le a}\,\mathrm{d}x\,\mathrm{d}y\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}}\,\mathrm{d}z=8\int_0^a\sqrt{a^2-x^2}\,\mathrm{d}x\int_0^{a-x}\,\mathrm{d}y=\left(2\pi-\frac{8}{3}\right)a^3.$$

3. 设 f 为连续函数. 求极限:

$$\lim_{r\to 0}\frac{3}{4\pi r^3}\iiint_{x^2+y^2+z^2\leqslant r^2} f(x,y,z)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z.$$

解. 由积分中值定理, 存在 (x_0,y_0,z_0) 满足 $x_0^2+y_0^2+z_0^2\le r^2$ 且

$$\frac{3}{4\pi r}\iiint_{x^2+y^2+z^2\le r^2} f(x,y,z)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z=\frac{3}{4\pi r}\cdot\frac{4}{3}\pi r^3 f\left(x_0,y_0,z_0\right)=f\left(x_0,y_0,z_0\right).$$

从而由 f 的连续性可知原极限为 $f(0,0,0)$.

4. 计算下列积分:

(1).
$$\iiint_{x^2+y^2+z^2\leqslant 2} \sqrt{x^2+y^2+z^2}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$

(2).
$$\iiint_D \left(x^2+y^2\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$

其中 $D=\left\{(x,y,z):z\geqslant 0, a^2\leqslant x^2+y^2+z^2\leqslant b^2\right\}$;

(3).
$$\iiint_D \left(1-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\right)\right)^{1/2}\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z,$$

其中 D 为椭球面 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ 包围的立体.

解.

(1). 做球坐标系换元

$$x=r\sin\theta\cos\varphi,\quad y=r\sin\theta\sin\varphi,\quad z=r\cos\theta,$$

从而

$$\iiint_{x^2+y^2+z^2\leq 2}\sqrt{x^2+y^2+z^2}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z=\int_0^{\sqrt{2}}\mathrm{d}r\int_0^{\pi}\mathrm{d}\theta\int_0^{2\pi}\mathrm{d}\varphi r^2\sin\theta\cdot r=4\pi.$$

(2). 做球坐标换元, 则

$$\begin{aligned}\iiint_D\left(x^2+y^2\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z&=\int_a^b\mathrm{d}r\int_0^{\pi/2}\mathrm{d}\theta\int_0^{2\pi}\mathrm{d}\varphi\cdot r^2\sin\theta\cdot\left((r\sin\theta\sin\varphi)^2+(r\sin\theta\cos\varphi)^2\right)\\&=\int_a^b r^4\,\mathrm{d}r\int_0^{\pi/2}\sin^3\theta\mathrm{d}\theta\int_0^{2\pi}\mathrm{d}\varphi=\frac{4\pi}{15}\left(b^5-a^5\right)\end{aligned}$$

(3). 做换元
$$x=ar\sin\theta\cos\varphi,\quad y=br\sin\theta\sin\varphi,\quad z=cr\cos\theta,$$

则

$$\iiint_D\left(1-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\right)^{1/2}\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z=\int_0^1\mathrm{d}r\int_0^{\pi}\mathrm{d}\theta\int_0^{2\pi}\mathrm{d}\varphi abcr^2\sin\theta\sqrt{1-r^2}=\frac{abc\pi^2}{4}.$$

5. 计算由下列曲面围成的立体的体积:

(1). $\left(x^2+y^2+z^2\right)^n=z^{2n-1}\,(n\in\mathbb{N}^*)$;

(2). $\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\right)^2=\frac{x^2}{a^2}+\frac{y^2}{b^2}$.

解.

(1). 我们有

$$\left(x^2+y^2+z^2\right)^n = z^{2n-1} \Longleftrightarrow x^2+y^2 = z^{2-1/n} - z^2.$$

从而

$$V = \int_0^1 \mathrm{d}z \int_{x^2+y^2\leq z^{2-1/n}-z^2} \mathrm{d}x\,\mathrm{d}y = \int_0^1 \pi\left(z^{2-1/n}-z^2\right)\mathrm{d}z = \frac{\pi}{9n-3}.$$

(2). 作换元 $u=ax, v=by, w=cz$ 后新的积分区域变为

$$u^2+v^2+w^2 = \sqrt{u^2+v^2}.$$

从而对一个确定的 w,

$$\frac{1-\sqrt{1-4w^2}}{2} \leq \sqrt{u^2+v^2} \leq \frac{1+\sqrt{1-4w^2}}{2}$$

其对应的面积为 $\pi\sqrt{1-4w^2}$, 于是

$$V = abc\int_{-1/2}^{1/2} \pi\sqrt{1-4w^2}\,\mathrm{d}w = \frac{\pi^2 abc}{4}$$

6. 计算积分

$$\int_{[0,1]^n} \cdots \int_1 (x_1+\cdots+x_n)^2\,\mathrm{d}x_1\cdots\mathrm{d}x_n$$

解. 我们有

$$I = \int_{[0,1]^n} \sum_{i,j=1}^n x_i x_j\,\mathrm{d}x_1\cdots\mathrm{d}x_n = \sum_{1\leq i\neq j\leq n} \iint_{[0,1]^2} x_i x_j\,\mathrm{d}x_i\,\mathrm{d}x_j + \sum_{i=1}^n \int_0^1 x_i^2\,\mathrm{d}x_i = \frac{n^2}{4}+\frac{n}{12}.$$

7. 计算累次积分:

$$\int_0^1 \mathrm{d}x_1 \int_0^{x_1} \mathrm{d}x_2 \cdots \int_0^{x_{n-1}} x_1\cdots x_{n-1}x_n\,\mathrm{d}x_n$$

解. 由第 12 题可知,

$$I = \frac{1}{n!}\left(\int_0^1 x\,\mathrm{d}x\right)^n = \frac{1}{2^n n!}$$

8. 计算下列 $\mathbb{R}^n$ 中区域的体积 $(a_1, a_2, \cdots, a_n > 0)$:

(1). $V_n = \left\{(x_1,x_2,\cdots,x_n) : \frac{x_1}{a_1}+\frac{x_2}{a_2}+\cdots+\frac{x_n}{a_n} \leqslant 1, x_1,x_2,\cdots,x_n \geqslant 0\right\}$;

(2). $V_n(a) = \{(x_1,x_2,\cdots,x_n) : |x_1|+|x_2|+\cdots+|x_n| \leqslant a\}$.

解.

(1). 做换元 $y_i = \frac{x_i}{a_i}$, 从而

$$V_n = \left(\prod_{i=1}^{n} a_i\right) \mu\left(D_n\right).$$

其中 $D_n := \{(x_1, \ldots, x_n) \mid x_i \geq 0, x_1 + \cdots + x_n \leq 1\}$. 从而

$$\begin{aligned}
\mu\left(D_n\right) &= \int_0^1 \mathrm{d}x_1 \int_0^{1-x_1} \mathrm{d}x_2 \cdots \int_0^{1-x_1-\cdots-x_{n-1}} \mathrm{d}x_n \\
&= \int_0^1 \mathrm{d}x_1 \int_0^{1-x_1} \mathrm{d}x_2 \cdots \int_0^{1-x_1-\cdots-x_{n-2}} \left(1 - x_1 - \cdots - x_{n-1}\right) \mathrm{d}x_{n-1}
\end{aligned}$$

从而根据对称性,

$$\begin{aligned}
\mu\left(D_n\right) &= \mu\left(D_{n-1}\right) - (n-1) \int_{D_{n-1}} x_1 \, \mathrm{d}x_1 \cdots \mathrm{d}x_{n-1} \\
&= \mu\left(D_{n-1}\right) - (n-1) \int_0^1 x_1 \, \mathrm{d}x_1 \int_0^{1-x_1} \mathrm{d}x_2 \cdots \int_0^{1-x_1-\cdots-x_{n-1}} \mathrm{d}x_{n-1} \\
&= \mu\left(D_{n-1}\right) - (n-1) \int_0^1 x_1 \left(1 - x_1\right)^{n-2} \mu\left(D_{n-2}\right) \mathrm{d}x_1 \\
&= \mu\left(D_{n-1}\right) - \frac{1}{n} \mu\left(D_{n-2}\right).
\end{aligned}$$

结合 $\mu\left(D_1\right) = 1, \mu\left(D_2\right) = \frac{1}{2}$, 递推可知 $D_n = \frac{1}{n!}$, 从而

$$V_n = \frac{1}{n!} \prod_{i=1}^{n} a_i$$

(2). 简单换元可知

$$V_n(a) = 2^n a^n \mu\left(D_n\right) = \frac{(2a)^n}{n!}.$$

9. 设 K 为二元连续函数，对 $n \in \mathbb{N}^*$ ，令

$$K_n(x, y) = \int_{[a,b]^n} \cdots \int_K K\left(x, t_1\right) K\left(t_1, t_2\right) \cdots K\left(t_n, y\right) \mathrm{d}t_1 \cdots \mathrm{d}t_n.$$

求证: 对任何 $m, n \in \mathbb{N}^*$, 有

$$K_{m+n+1}(x, y) = \int_a^b K_m(x, t) K_n(t, y) \mathrm{d}t$$

证明. 我们有

$$\begin{aligned}
K_{n+m+1}(x, y) &= \int_{[a,b]^{n+m+1}} K\left(x, t_1\right) \cdots K\left(t_n, t_{n+1}\right) K\left(t_{n+1}, t_{n+2}\right) \cdots K\left(t_{n+m+1}, y\right) \mathrm{d}t_1 \cdots \mathrm{d}t_{n+m+1} \\
&= \int_a^b \mathrm{d}t_{n+1} \left(\int_{[a,b]^n} K\left(x, t_1\right) \cdots K\left(t_n, t_{n+1}\right) \mathrm{d}t_1 \cdots \mathrm{d}t_n\right) \\
&\qquad \left(\int_{[a,b]^m} K\left(t_{n+1}, t_{n+2}\right) \cdots K\left(t_{n+m+1}, y\right) \mathrm{d}t_{n+2} \cdots \mathrm{d}t_{n+m+2}\right) \\
&= \int_a^b K_n\left(x, t_{n+1}\right) K_m\left(t_{n+1}, y\right) \mathrm{d}t_{n+1}.
\end{aligned}$$

□

10. 设 $a_1, a_2, \cdots, a_n > 0$,

$$V_n = \left\{ (x_1, x_2, \cdots, x_n) : \frac{|x_i|}{a_i} + \frac{|x_n|}{a_n} \leqslant 1 (i = 1, 2, \cdots, n-1) \right\}.$$

求 V_n 的体积.

解. 做换元 $x_i = a_i y_i$, 根据对称性和 Fubini 定理, 我们有

$$\begin{aligned} \mu(V_n) &= 2^n \left(\prod_{i=1}^n a_i \right) \int_0^1 \mathrm{d}x_n \int_0^{1-x_n} \mathrm{d}x_1 \cdots \int_0^{1-x_n} \mathrm{d}x_{n-1} \\ &= 2^n \left(\prod_{i=1}^n a_i \right) \int_0^1 (1-x_n)^{n-1} \, \mathrm{d}x_n = \frac{2^n}{n} \prod_{i=1}^n a_i. \end{aligned}$$

11. 设 f 是单变量连续函数. 证明:

$$\int_0^a \mathrm{d}x_1 \int_0^{x_1} \mathrm{d}x_2 \cdots \int_0^{x_{n-1}} f(x_n) \, \mathrm{d}x_n = \frac{1}{(n-1)!} \int_0^a f(t)(a-t)^{n-1} \, \mathrm{d}t.$$

证明. 注意到等式左侧即 $f(x_n)$ 在

$$D := \{ (x_1, \ldots, x_n) \mid 0 \leq x_n \leq \cdots \leq x_1 \leq a \}$$

上的积分值. 从而根据 Fubini 定理,

$$\text{LHS} = \int_0^a f(x_n) \, \mathrm{d}x_n \int_{x_n}^a \mathrm{d}x_{n-1} \cdots \int_{x_2}^a \mathrm{d}x_1.$$

根据第 12 题的结论, 这就是

$$\text{LHS} = \int_0^a f(x_n) \, \mathrm{d}x_n \frac{1}{(n-1)!} (a - x_n)^{n-1} = \text{RHS}.$$

□

12. 设 f 是单变量连续函数. 证明:

$$\int_0^a \mathrm{d}x_1 \int_0^{x_1} \mathrm{d}x_2 \cdots \int_0^{x_{n-1}} f(x_1) f(x_2) \cdots f(x_n) \, \mathrm{d}x_n = \frac{1}{n!} \left(\int_0^a f(t) \mathrm{d}t \right)^n.$$

证明. 考虑 $D := [0, a]^n$, 我们用所有 $n!$ 个 n 阶排列 p 将其划分为 $n!$ 个区域 D_p 满足

$$D_p = \left\{ (x_1, \ldots, x_n) \mid 0 \leq x_{p_1} \leq \cdots \leq x_{p_n} \leq a \right\}$$

从而 D_p 两两不交且 $D = \bigcup D_p$. 于是

$$\begin{aligned} \left(\int_0^a f(t) \mathrm{d}t \right)^n &= \int_D f(x_1) \cdots f(x_n) \, \mathrm{d}x_1 \cdots \mathrm{d}x_n \\ &= \sum_{p \in \mathfrak{E}_n} \int_{D_p} f(x_1) \cdots f(x_n) \, \mathrm{d}x_1 \cdots \mathrm{d}x_n \\ &= \sum_{p \in \mathfrak{E}_n} \int_0^a f(x_{p_n}) \, \mathrm{d}x_{p_n} \int_0^{x_{p_n}} f(x_{p_{n-1}}) \, \mathrm{d}x_{p_{n-1}} \cdots \int_0^{x_{p_2}} f(x_{p_1}) \, \mathrm{d}x_{p_1} \\ &= n! \int_0^a \mathrm{d}x_1 \int_0^{x_1} \mathrm{d}x_2 \cdots \int_0^{x_{n-1}} f(x_1) \cdots f(x_n) \, \mathrm{d}x_n \end{aligned}$$

□

13. 设 f 为 n 元连续函数. 证明:

$$\int_a^b \mathrm{d}x_1 \int_a^{x_1} \mathrm{d}x_2 \cdots \int_a^{x_{n-1}} f(x_1, x_2, \cdots, x_n)\, \mathrm{d}x_n = \int_a^b \mathrm{d}x_n \int_{x_n}^b \mathrm{d}x_{n-1} \cdots \int_{x_2}^b f(x_1, x_2, \cdots, x_n)\, \mathrm{d}x_1$$

证明. 考虑区域

$$D := \{(x_1, \ldots, x_n) \mid a \le x_n \le \cdots \le x_1 \le b\}$$

则根据 Fubini 定理,

$$\int_a^b \mathrm{d}x_1 \int_a^{x_1} \mathrm{d}x_2 \cdots \int_a^{x_{n-1}} f(x_1, \ldots, x_n)\, \mathrm{d}x_n = \int_D f(\mathbf{x}) \mathrm{d}\mathbf{x} = \int_a^b \mathrm{d}x_n \int_{x_n}^b \mathrm{d}x_{n-1} \cdots \int_{x_2}^b f(x_1, \ldots, x_n)\, \mathrm{d}x_1.$$

□